9.5 Arc Length and Sector Area

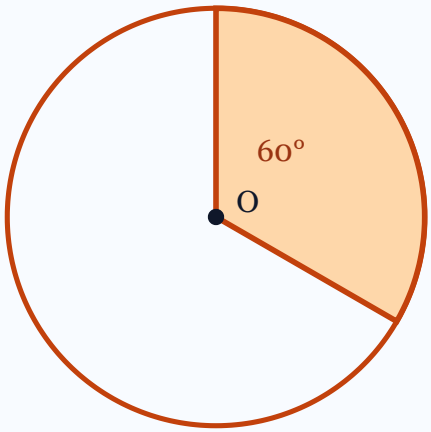
A central angle θ claims the fraction $\dfrac{\theta}{360^\circ}$ of an entire circle. Apply that fraction to the circle's circumference to get **arc length**, $\ell = \dfrac{\theta}{360^\circ} \cdot 2\pi r$, or to the circle's area to get **sector area**, $A = \dfrac{\theta}{360^\circ} \cdot \pi r^2$. A useful shortcut follows directly: since both formulas share the same fraction, $A = \dfrac{1}{2} r\ell$ — a sector's area is half its radius times its arc length, just like a triangle's area is half its base times its height. This worksheet leaves answers in terms of π when the numbers work out exactly, and uses the arc–area shortcut to solve for two unknowns at once.

Examples

Example 1: Arc length

A circle has radius 9 and central angle 60°. Find the arc length.

Solution



The sector claims $\frac{60}{360} = \frac{1}{6}$ of the full circle.

Identify the fraction of the circle: $\dfrac{60^\circ}{360^\circ} = \dfrac{1}{6}$.

Apply that fraction to the full circumference $2\pi r$: $\ell = \dfrac{1}{6} \cdot 2\pi(9) = \dfrac{1}{6}(18\pi) = 3\pi$.

Answer: 3π.

Example 2: Sector area

A circle has radius 12 and central angle $90°$. Find the sector area.

Solution

Identify the fraction of the circle: $\dfrac{90°}{360°} = \dfrac{1}{4}$.

Apply that fraction to the full area πr^2: $A = \dfrac{1}{4} \cdot \pi(12)^2 = \dfrac{1}{4}(144\pi) = 36\pi$.

Answer: 36π.

Example 3: Angle from arc length

A circle has radius 8 and arc length 4π. Find θ.

Solution

Identify: this reverses the arc length formula — solve $\ell = \dfrac{\theta}{360} \cdot 2\pi r$ for θ instead of ℓ.

Set up the equation: $4\pi = \dfrac{\theta}{360} \cdot 2\pi(8) = \dfrac{\theta}{360} \cdot 16\pi$.

Solve: $\dfrac{\theta}{360} = \dfrac{4\pi}{16\pi} = \dfrac{1}{4}$, so $\theta = \dfrac{1}{4}(360°) = 90°$.

Answer: $90°$.

Example 4: Radius from sector area

A sector has area 24π and central angle $120°$. Find the radius.

Solution

Identify: solve the sector area formula $A = \dfrac{\theta}{360}\pi r^2$ for r this time.

Set up the equation: $24\pi = \dfrac{120}{360}\pi r^2 = \dfrac{1}{3}\pi r^2$.

Solve: $r^2 = 3(24) = 72$, so $r = \sqrt{72} = \sqrt{36 \cdot 2} = 6\sqrt{2}$.

Answer: $6\sqrt{2}$.

Example 5: Pizza slice

A pizza has radius 7 in and a slice with central angle 45°. Find the length of crust (arc) on the slice, and its area.

Solution

Crust length: $\ell = \dfrac{45}{360} \cdot 2\pi(7) = \dfrac{1}{8}(14\pi) = \dfrac{7\pi}{4}$.

Area: $A = \dfrac{45}{360} \cdot \pi(7)^2 = \dfrac{1}{8}(49\pi) = \dfrac{49\pi}{8}$.

Answer: crust $= \dfrac{7\pi}{4}$ in; area $= \dfrac{49\pi}{8}$ in^2.

Example 6: Arc length

Radius 6, angle 120°. Find the arc length.

Solution

Identify the fraction of the circle: $\dfrac{120^\circ}{360^\circ} = \dfrac{1}{3}$. Full circumference: $2\pi(6) = 12\pi$.

Compute: $\dfrac{1}{3} \cdot 12\pi = 4\pi$.

Answer: 4π.

Example 7: Sector area

Radius 10, angle 72°. Find the sector area.

Solution

Identify the fraction of the circle: $\dfrac{72^\circ}{360^\circ} = \dfrac{1}{5}$. Full area: $\pi(10)^2 = 100\pi$.

Compute: $\dfrac{1}{5} \cdot 100\pi = 20\pi$.

Answer: 20π.

Example 8: Find the angle

Radius 15, arc length 5π. Find θ.

Solution

Identify: solve the arc length formula for θ, using the full circumference $2\pi(15) = 30\pi$.

Set up and solve: $5\pi = \dfrac{\theta}{360} \cdot 30\pi \Rightarrow \dfrac{\theta}{360} = \dfrac{5\pi}{30\pi} = \dfrac{1}{6} \Rightarrow \theta = 60°$.

Answer: $60°$.

Example 9: Find the radius

Sector area 27π, angle $270°$. Find r.

Solution

Identify: solve the sector area formula for r, using the given angle to get the fraction $\dfrac{270}{360} = \dfrac{3}{4}$.

Set up and solve: $27\pi = \dfrac{3}{4}\pi r^2 \Rightarrow r^2 = 27 \cdot \dfrac{4}{3} = 36 \Rightarrow r = \sqrt{36} = 6$.

Answer: 6.

Example 10: Find both radius and angle from arc length and area

A sector has arc length 6π and area 27π. Find the radius and the central angle.

Solution

Use the shortcut $A = \dfrac{1}{2}r\ell$:

$27\pi = \dfrac{1}{2}r(6\pi) = 3\pi r$

$r = 9$.

Now find θ from the arc length: $6\pi = \dfrac{\theta}{360} \cdot 2\pi(9) = \dfrac{\theta}{360} \cdot 18\pi$

$\dfrac{\theta}{360} = \dfrac{1}{3} \Rightarrow \theta = 120°$.

Answer: $r = 9$; $\theta = 120°$.

Example 11: Application — windshield wiper

A windshield wiper blade is 24 in long (pivoting at one end) and sweeps through an angle of $120°$. Find the area it cleans and the length of the arc traced by the tip.

Solution

Area (sector): $A = \frac{120}{360} \cdot \pi(24)^2 = \frac{1}{3}(576\pi) = 192\pi$ in^2.

Arc length (tip's path): $\ell = \frac{120}{360} \cdot 2\pi(24) = \frac{1}{3}(48\pi) = 16\pi$ in.

Answer: area $= 192\pi$ in^2; arc length $= 16\pi$ in.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Arc length

Radius 4, angle 90°. Find the arc length.

Answer: **Verification:**

Problem 2: Sector area

Radius 8, angle 45°. Find the sector area.

Answer: **Verification:**

Problem 3: Arc length

Radius 10, angle 180°. Find the arc length.

Answer: **Verification:**

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Problem 4: Sector area

Radius 6, angle 60°. Find the sector area.

Answer: **Verification:**

Problem 5: Find angle

Radius 9, arc length 3π. Find θ.

Answer: **Verification:**

Problem 6: Find radius

Sector area 12π, angle 90°. Find r.

Answer: **Verification:**

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Problem 7: Arc and area

Radius 4, angle 150°. Find the arc length and sector area.

Answer: **Verification:**

Problem 8: Arc length

Radius 12, angle 30°. Find the arc length.

Answer: **Verification:**

Problem 9: Sector area

Radius 5, angle 144°. Find the sector area.

Answer: **Verification:**

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Problem 10: Find angle

Radius 6, arc length 2π. Find θ.

Answer: **Verification:**

Problem 11: Find radius

Arc length 4π, angle $120°$. Find r.

Answer: **Verification:**

Problem 12: Semicircle sector

Radius 7, angle $180°$. Find the sector area.

Answer: **Verification:**

Problem 13: Fraction of circle

What fraction of a circle is a $40°$ sector?

Answer: **Verification:**

Problem 14: Challenge

A sector has arc length 6π and area 30π. Find the radius and the central angle.

Answer: **Verification:**

Problem 15: Challenge

A rotating sprinkler sprays water out to 15 ft and rotates through $150°$. Find the area watered (leave in terms of π, then approximate using $\pi \approx 3.14$) and the arc length of the spray's edge (in terms of π).

Answer: **Verification:**

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✓ Answer Key

1. 2π
2. 8π
3. 10π
4. 6π
5. $60°$
6. $4\sqrt{3}$
7. Arc $\frac{10\pi}{3}$; area $\frac{20\pi}{3}$
8. 2π
9. 10π
10. $60°$
11. $r = 6$
12. $\frac{49\pi}{2}$
13. $\frac{1}{9}$
14. $r = 10$; $\theta = 108°$
15. Area $= 93.75\pi \approx 294$ ft^2; arc length $= 12.5\pi$ ft

Great work — keep practising!